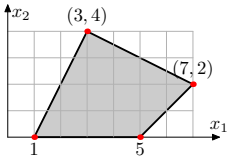
07-LP-exc

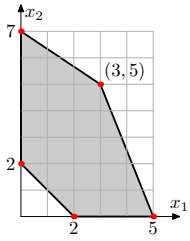
Geometric descriptions & solutions in $\mathbb{R}^2$

1. Define the shaded region 

by system of inequalities: $\begin{cases} a_{11}x_1 + a_{12}x_2 \le b_1, \\ \vdots \\ a_{41}x_1 + a_{42}x_2 \le b_4. \end{cases}$ 5b6ab6

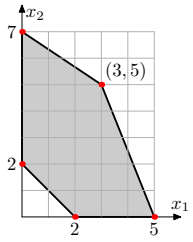
2. Describe the shaded plane region to the right by a system of inequalities: d7b56d

$$\begin{cases} a_{11}x_1 + a_{12}x_2 \le b_1, \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 \le b_m. \end{cases}$$



3. Describe the shaded plane region D to the right by a system of inequalities: d7b56e

$$D: \begin{cases} a_{11}x_1 + a_{12}x_2 \le b_1, \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 \le b_m. \end{cases}$$

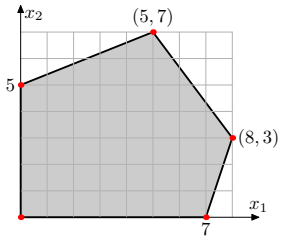


Find both the max and min value of the function $z = 3x_1 + 4x_2$ for $(x_1, x_2) \in D$.

4. Define by inequalities the finite region D bounded by the convex quadrilateral with vertices at (0,3), (3,0), (7,5) and (5,7), sides and vertices included. Find afterwards graphically the max of the objective function e38a6d

$$z = 7x_1 + 10x_2,\ (x_1, x_2) \in D.$$

5. Define by system of inequalities $A\mathbf{x} \le \mathbf{b}$ the shaded region Ω below: 30ace7



$\rightsquigarrow \quad A\mathbf{x} \le \mathbf{b} \Leftrightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 \le b_1, \\ a_{21}x_1 + a_{22}x_2 \le b_2, \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 \le b_m. \end{cases}$

6. Formulate the *dual* of the LP to the right and solve it *graphically*, by: $\min w = 21y_1 + 41y_2 + 25y_3$ 30ace9

$$\begin{cases} 3y_1 + 4y_2 - 2y_3 \ge 3 \\ -y_1 + 3y_2 + 5y_3 \ge 2 \\ y_1, y_2, y_3 \ge 0 \end{cases}$$

   (a) sketching the feasible region Ω of the dual LP; and

   (b) finding the corner point of Ω at which the objective function of the dual obtains its optimum.

7. Formulate the *dual* of the LP to the right and solve it either $\min w = 5y_1 + 8y_2 + 10y_3$ 91cff0

$$\text{s.t.} \begin{cases} y_1 + y_2 + 2y_3 \ge 3 \\ y_1 + 2y_2 + y_3 \ge 2 \\ y_1, y_2, y_3 \ge 0 \end{cases}$$

   (a) graphically, by sketching the feasible region Ω of the dual LP; and then finding the corner point at which the opt is attained, or

   (b) using the Simplex Method.

Problem def

1. FuelCo blends two types of gasoline, Blend 1 and Blend 2. Each blend reguires ingredients A, B and C in 8b1191

proportions as indicated below. The available quantities (per timeslice) for the three ingrediens in gallons as well as the corresponding profit in \$/gallon for Blend 1 and Blend 2 are as follows:

	Blend 1	Blend 2	Available
A	1/6	1/4	60
B	1/2	1/4	60
C	1/3	1/2	90
Profit	3	4	

Formulate a LP-problem to maximize FuelCo's profit and solve it by the Simplex method.

2. SportsCo plans a local TV-ad campaign for tennis equipment in conjunction with a tennis tournament held in town during the weekend. One can choose between day-time advertising, evening advertising and match-time (Sunday afternoon) advertising. Price info is according to the table below: 2f7b25

Day	Time	Price per slot	Estimated Audience
Friday	day-time	\$6000	3500
	evening	\$9000	5000
Saturday	day-time	\$7500	5000
	evening	\$9000	5000
Sunday	match-time	\$45000	30000
	evening	\$9000	5000

Formulate the LP (no solving required) minimizing the campaign cost, if:

- The campaign should reach at least an audience of at least 350000;
- The total number of allocated slots does not exceed 25;
- At least 6 slots allocated each day, the Sunday ones being at most 9;
- At least half of the slots are match-time.

3. PlastCo is optimizing the production of three plastic items undergoing two processing phases. The time in sec/unit required for each of the parts as well as the time available during one production cycle are given in the table to the right. 512d7a

	Item 1	Item 2	Item 3	Time
Phase 1	3	1	2	150
Phase 2	5	2	3	180
Profit	24	12	18	

(a) Help PlastCo to formulate an LP-problem to maximize the profit per cycle.

(b) Find the optimal plan using the simplex method. Is the solution unique?

Standard Simplex Tables or Geom. solutions

1. Use the Simplex Method to find the optimal solution $\mathbf{z}^* = z(\mathbf{x}^*)$ of the LP below: f1ae65

$$\max \quad z = 2x_1 + 26x_2 + 19x_3, \ \mathbf{x} \geq \mathbf{0},$$
$$\text{s. t.} \quad \begin{bmatrix} 1 & 4 & 3 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 23 \\ 16 \end{bmatrix}$$

Give the point $\mathbf{x}^*$ this max is attained, as well as which of the constraints are active at optimum.

2. Solve by the simplex method the LP (include all the Simplex tableaux in your solution): 4421d4

$$\max \quad z = 3x_1 + 2x_2 + x_3$$
$$\text{s.t.} \quad \begin{cases} x_1 + x_2 + x_3 & \leq 120 \\ 2x_1 + 3x_2 - x_3 & \leq 90 \\ x_1, x_2, x_3 & \geq 0. \end{cases}$$

3. Solve by the Simplex method the max LP to the right. State the optimal value z^*, the point x^* at which it is attained, as well as which constraints are active at optimum. In addition, write down explicitly the objective function z^* as a function of the non-basic variables at optimum.

$$\max \quad z = \begin{bmatrix} 8 & 2 & 7 \end{bmatrix} \mathbf{x}, \quad \mathbf{x} \geq \mathbf{0}, \qquad \text{s.t.} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 25 \\ 2 \\ 12 \end{bmatrix}$$

51a405

4. Solve LP Problem:

$$\begin{array}{rrcrcrcl} \max\ z = & 3x_1 & + & 4x_2, & \mathbf{x} \geq \mathbf{0} \\ \textbf{s.t.} & x_1/5 & + & x_2/4 & \leq & 60 \\ & x_1/2 & + & x_2/4 & \leq & 60 \\ & x_1/3 & + & x_2/2 & \leq & 90 \end{array}$$

5. Use the Simplex Method to solve the LP:

$$\begin{array}{rcrcrcrcl} \max z & = & 4x_1 & + & x_2 & + & 11x_3, & \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & & 3x_1 & + & x_2 & + & 7x_3 & \leq & 72, \\ & & 2x_1 & + & x_2 & + & 6x_3 & \leq & 60. \end{array}$$

484da3

6. Use the Simplex Method to solve the LP:

df8ef0

$$\begin{array}{rcrcrcrcl} \min z & = & -2x_1 & - & 4x_2 & + & 3x_3 & & \\ \text{s.t.} & & 2x_1 & + & 4x_2 & + & 3x_3 & \leq & 100 \\ & & x_1 & + & 3x_2 & + & 2x_3 & \leq & 60 \\ & & x_1 \geq 0, & & x_2 \geq 0, & & x_3 \geq 0. & & \end{array}$$

Is the solution unique?

7. Resolve graphically or via the Simplex method the LP:

$$\begin{cases} \max & z = \begin{bmatrix} 1 & 2 \end{bmatrix} \mathbf{x}, \ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 60 \\ 140 \\ 105 \end{bmatrix}. \end{cases}$$

9ae85b

8. The meant-to-be correct Simplex Tableau below, correctly computed by a meticulous student from the canonical representation of a max-LP problem, did not end up being entirely correct.

af0a9f

	z	x_1	x_2	x_3	s_1	s_2	s_3	$\hat{\mathbf{b}}$
z	1	3	0	0	2	2	0	16
s_1	0	-2	1/2	0	1	$-3/2$	0	1/2
x_3	0	1	1/2	1	0	1/2	0	3/2
s_3	0	7	3	1	0	1	1	6

Take over and make the necessary row operations to produce a legal Simplex Tableau with the basis variables indicated in the leftmost column. Complete then the Simplex algotithm to find the optimal value, max z^*.

9. Solve the following LP using the Simplex method. State the optimal solution z^* and the point $\mathbf{x}^*$ at which it is attained.

de76bc

$$\begin{array}{rrcrcrcl} \max & 12x_1 & + & 20x_2 & + & 15x_3, & \mathbf{x} \geq \mathbf{0} \\ \text{s. t.} & 2x_1 & + & 5x_2 & + & 3x_3 & \leq & 50 \\ & x_1 & + & x_2 & + & x_3 & \leq & 12 \\ & & & x_2 & & & \leq & 5 \end{array}$$

10. Use the Simplex Method to solve the LP:

0e1003

$$\min \ z = 5x_1 - 8x_2 - 3x_3, \ \mathbf{x} \geq \mathbf{0}$$
$$\text{s.t.} \begin{cases} 2x_1 + 5x_2 - x_3 & \leq 1 \\ -3x_1 - 8x_2 + 2x_3 & \leq 4 \\ -2x_1 - 12x_2 + 3x_3 & \leq 9 \end{cases}$$

11. The LP problem attains its max at $\mathbf{x}^* = (x_1, 0, 0, 0)$, $x_1 > 0$. ca94b4

$$\begin{aligned} \max \quad & z = \begin{bmatrix} 6 & c_2 & c_3 & c_4 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} \quad & \begin{bmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \leq \begin{bmatrix} 5 \\ 3 \end{bmatrix} \end{aligned}$$

(a) Solve the problem, ie., find x_1. Which constraints are then active at optimum?

(b) In which ranges can the coefficients c_2, c_3 and c_4 vary without change of basis occurs?

Matrices in LP $(BN|\mathbf{b})$, B^{-1}, etc.

1. Find the optimal solution z^* for the LP below as well as at which $\mathbf{x}^*$ is it attained 5450f8

$$\begin{aligned} \max \quad & z = \begin{bmatrix} 6 & 6 & 1 & 4 & 0 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} \quad & \begin{bmatrix} 4 & -1 & -1 & 3 & 1 \\ 1 & 2 & 1 & 2 & -1 \\ -1 & 1 & 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 14 \\ 12 \\ 10 \end{bmatrix} \end{aligned}$$

if the basic variables at optimum are x_1, x_2 and x_5. Deduce also which ones of the three constraints are binding.

2. Find the optimal solution z^* for the LP below as well as at which $\mathbf{x}^*$ is it attained a9cf00

$$\begin{cases} \max \quad z = \begin{bmatrix} 3 & 1 & 2 & 5 & 4 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} \begin{bmatrix} 3 & 1 & 2 & 1 & 4 \\ 1 & 5 & 2 & 1 & 2 \\ 7 & 2 & 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 1200 \\ 2400 \\ 1800 \end{bmatrix}, \end{cases}$$

if one knows that at optimum, x_3 and x_4 are the basic variables and only constraints 1 and 3 are binding.

3. A normal max LP (P): $\{\max\ z = \mathbf{c}^T\mathbf{x},\ A\mathbf{x} \leq \mathbf{b}\}$, is solved with final Simplex Tableau as follows: 32d008

$$(P): \begin{cases} \max\ z = \mathbf{c}^T\mathbf{x} \\ A\mathbf{x} \leq \mathbf{b} \end{cases} \overset{\text{Splx}}{\rightsquigarrow}$$

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	rhs
z	1	0	3	4	0	1	3	?
x_5	3	0	1	−1	1	5	−7	4
x_2	2	1	4	3	0	−2	3	3

Let (D):$\{\min\ w = b^T\mathbf{y},\ A^T\mathbf{y} \geq \mathbf{c}\}$ be the corresponding dual LP of (P).

(a) Find the intensity matrix A and the capacities $\mathbf{b}$ for the constraints of (P).

(b) Find the optimal points $\mathbf{x}^*$ and $\mathbf{y}^*$ as well as the common optimum $z^* = \mathbf{y}^{*T}A\mathbf{x}^* = w^*$.

(c) (Optional) State the full original formulation of both the primal (P) and the dual (D).

4. The LP problem below has x_1 and x_2 as basic variables at optimum: 2cf83b

$$\begin{aligned} \max \quad & z = \begin{bmatrix} -1 & 9 & c_3 & c_4 \end{bmatrix} \mathbf{x}, \quad \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} \quad & \begin{bmatrix} 1 & 1 & 3 & 4 \\ -3 & 2 & -5 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 1 \end{bmatrix} \end{aligned}$$

(a) Find the point $\mathbf{x}^*$ at which the optimum $z^* = z(x^*)$ is attained. State also the value of z^*.

(b) What are the shadow prices $\mathbf{y}^{*T} = \nabla_{\mathbf{b}} z^*$ at optimum?

(c) Determine the greatest values of c_3 and c_4 can get before a change of basis becomes necessary.

Big M

1. Use the Big-M method to rewrite in standard form and compose the initial Simplex Tableau for the LP: 5681b6

$$\begin{array}{lrcrcrcl} \max & z & = & x_1 & + & 3x_2 & & \\ \text{s.t.} & & & 2x_1 & + & 3x_2 & \geq & 24 \\ & & & x_1 & - & x_2 & \leq & 7 \\ & & & & & x_2 & \leq & 6 \end{array}$$

Make the necessary row operations to obtain a starting Tableau with initial feasible basic solution. Perform then the first step in the Simplex algorithm and produce the corresponding Tableau. (Completing all the steps until optimum is reached is a feather in the hat, but is *not required* for full score).

2. Use the Big-M method to rewrite in standard form and compose the initial Simplex Tableau for the LP to the right. As a first step, obtain a feasible initial basic solution. Proceed with one step in the Simplex algorithm and just mark the pivot element for the nextcoming step in the Tableau. Finding optimum is a merit but is not required. f86afd

$$\max \quad z = \begin{bmatrix} 5 & 3 & 6 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0},$$
$$\text{s.t.} \quad \begin{bmatrix} 2 & -3 & 5 \\ 4 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{matrix} \geq \\ \leq \end{matrix} \begin{bmatrix} 30 \\ 240 \end{bmatrix}$$

3. Use the Big-M method to find an initial basic feasible solution and then solve the LP-problem: f32d2a

$$\begin{array}{lrcrcrcrcl} \text{s.t.}\max & z & = & 3x_1 & + & 4x_2 & + & 2x_3 & & \\ & & & x_1 & + & x_2 & + & x_3 & \leq & 6 \\ & & & & - & x_2 & + & x_3 & \geq & 2 \\ & & & x_1 \geq 0, & & x_2 \geq 0, & & x_3 \geq 0. & & \end{array}$$

4. Use the Big-M method to rewrite in standard form and compose the initial Simplex Tableau for the LP: fa0e75

$$\begin{array}{rrcrcrcrl} \max & z = & 3x_1 & + & 2x_2 & + & x_3, & \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & & x_1 & + & 3x_2 & + & 8x_3 & \leq 72, \\ & & x_1 & + & 2x_2 & + & 5x_3 & \geq 12. \end{array}$$

Make the necessary row operations to obtain a starting Tableau with initial feasible basic solution. Perform then the first step in the Simplex algorithm and produce the corresponding Tableau. (Completing of all the steps until optimum is reached is a merit, but is not required for full score).

5. Solve the LP problem using the Big-M method: 0fa5f6

$$\begin{array}{rrcrcrcrl} \max & z & = & 2x_1 & + & 3x_2 & + & 4x_3, & \mathbf{x} \geq \mathbf{0} \\ \text{s.t.} & & & x_1 & + & 2x_2 & + & x_3 & \leq 4 \\ & & & 2x_1 & + & 3x_2 & + & 3x_3 & \geq 1 \end{array}$$

6. Use the Big-M method to solve the LP-problem: 952c1f

$$\begin{array}{rrcrcrcrl} \max & z = & -x_1 & + & x_2 & - & x_3, & \mathbf{x} \geq \mathbf{0} \\ \text{s.t.} & & 2x_1 & + & 2x_2 & + & x_3 & \leq 10 \\ & & x_1 & & & + & 2x_3 & \geq 5 \end{array}$$

Big-M with equality

1. The LP is solved by the Simplex method with final Tableau to the right: 674cc4

$$\max \ z = x_1 + 2x_2 + 3x_3,\ x_{1,2,3} \geq \mathbf{0}$$
$$\text{s.t.} \begin{cases} x_1 + x_2 + 2x_3 \geq 25 \\ 2x_1 + x_2 + x_3 \leq 50 \\ x_1 - x_2 + x_3 = 30 \end{cases}$$

$\overset{\text{Splx}}{\rightsquigarrow}$

z	9/2	0	0	0	5/2	1/2	140
x_3	3/2	0	1	0	1/2	1/2	40
x_2	1/2	1	0	0	1/2	−1/2	10
s_1	5/2	0	0	1	3/2	1/2	65

(a) State the optimum $z^* = z(\mathbf{x}^*)$, $\mathbf{x}^*$ at which it is attained, and which constraints active at opt.

(b) Now formulate the Dual of the LP above and answer the same questions as in (a)!

2. Introduce slack $s_1 \geq 0$ also for the equality constraint in the LP to the right. Then, Big-M penalize it as an artificial variable in the objective function to ensure it cannot be basic at optimum. Find that optimal value by the Simplex method. State as well the objective function z^* as a function of the non-basic variables at optimum.

$$\max\ z = 8x_1 + 2x_2 + 7x_3,\ \mathbf{x} \geq \mathbf{0}, \qquad \text{s.t.} \begin{cases} x_1 + 2x_2 + 3x_3 &= 25 \\ 4x_1 \quad\quad + x_3 &\leq 2 \\ -x_1 + x_2 + 2x_3 &\leq 12 \end{cases}$$

51a406

3. Introduce slack $s_1 \geq 0$ also for the equality constraint in the LP to the right. Then, Big-M penalize it as an artificial variable in the objective function to ensure it cannot be basic at optimum. Find that optimal value by the Simplex method. State as well the objective function z^* as a function of the non-basic variables at optimum.

$$\max\ z = \begin{bmatrix} 32 & 6 & 12 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \qquad \text{s.t.} \begin{bmatrix} 2 & 0 & 1 \\ 4 & 1 & 1 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{matrix} = \\ \leq \\ \leq \end{matrix} \begin{bmatrix} 8 \\ 16 \\ 51 \end{bmatrix}$$

bb2973

Rip info from glpk

1. Six processes apply for cycles $(x_1, \ldots, x_6)$ from the CPU for the next time slice. The scheduler tries to minimize its load z while abiding priorities/weights of processes and working under the restriction from the LP below. A sample output of a LP solver (`glpk`) is provided to the right.

2f7b26

$$\min\ z = \begin{bmatrix} 6 & 9 & 8 & 9 & 45 & 9 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0},$$

$$\text{s.t.} \begin{bmatrix} 7 & 5 & 12 & 9 & 30 & 5 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} \begin{matrix} \leq \\ \leq \\ \leq \\ \leq \\ \geq \\ \geq \\ \geq \end{matrix} \begin{bmatrix} 350 \\ 25 \\ 9 \\ 5 \\ 6 \\ 6 \\ 6 \end{bmatrix}$$

```
Zopt = 399|Xopt: 6|Reduced 0|Shadow   0|
          |      0|costs:  3|prices:  0|
          |      6|        0|         0|
          |      0|        1|       -45|
          |      7|        0|        51|
          |      0|       54|        53|
          |       |         |         0|
```

(a) Translate the solver output by writing z^* at optimum as a function of the non-basic variables.

(b) Which processes are halted at opt and how many cycles obtain the active ones? The load coefficient c_k of which sleeping process must drop the least so that it becomes re-activated?

(c) Which of the constraints are binding (sharp) at opt and which are not? By how much would the min value z^* drop if right hand sides of each constraint were *relaxed*, one at a time, by 1? (Specify exact the new value of the corresponding rhs b_i, as well as the amount of change of z^*).

Duality

1. Formulate the dual of the max LP to the right. Resolve it graphically and then use the solution to solve the primal problem. State explicitly the point x^* at which the optimal value z^* is attained.

$$\max\ z = \begin{bmatrix} 1 & 2 & -2 & 1 & -1 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \qquad \text{s.t.} \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \begin{matrix} \leq \\ = \end{matrix} \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

8c2aea

2. The LP to the right has an optimal solution $z^* = 22$ attained at $\mathbf{x}^* = (0, 0, 10, 3)^T$.

   (a) Formulate the dual problem.

   (b) Use the solution of the original problem to solve dual.

$$\max\ z = 3x_1 + 2x_2 + x_3 + 4x_4 \qquad \text{s.t.} \begin{cases} x_1 + x_2 + x_3 - x_4 \leq 7 \\ 2x_1 + 3x_2 + 5x_3 + x_4 \geq 20 \\ x_1 \quad\quad + x_4 \leq 3 \\ x_1 \geq 0,\ x_2 \leq 0,\ x_{3,4} \geq 0 \end{cases}$$

cdaca8

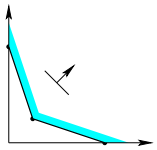
3. A farmer who raises chickens tries to find the cheapest daily diet which meets certain minimal nutritional requirements. The relevant data is summarized in the table below:

0fad69

Nutrient	Corn	Lime	Alfalfa	Minimal daily intake
Protein	8	4	4	10
Carbohydrates	4	2	4	6
Vitamines	2	3	4	5
Unit cost	15	8	5	

Formulated as a min LP problem, the solution suggests use of automatic variables. Instead, the farmer formulates the dual LP and obtains the solution in just one iteration of the Simplex method. What cost does the optimal diet have and how much of daily proteins, carbohydrates and vitamins does it provide?

4. Formulate and solve by the Simplex method the dual (D) of the example (P) from the BigM-lecture: 42fc8e



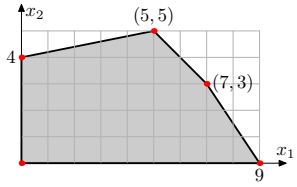
$$(P): \quad \min z = x_1 + x_2, \ \mathbf{x} \geq 0, \quad \text{s.t.} \begin{cases} 2x_1 + x_2 \geq 18 \\ x_1 + 2x_2 \geq 12 \end{cases}$$

Deduce afterwards from (D)'s final Tableau the solution of (P), $z^* = z(x^*)$.

5. Resolve the LP by formulating and solving its dual: 9ae85c

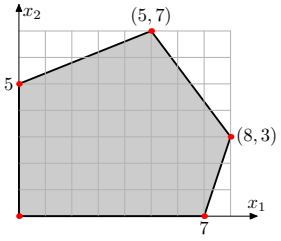
$$\begin{array}{rrcrcrcl} \min w = & 60y_1 & + & 140y_2 & + & 105y_3, & \mathbf{y} \geq 0, \\ \text{s.t.} & y_1 & + & 3y_2 & + & 2y_3 & \geq & 1, \\ & 2y_1 & + & 4y_2 & + & 5y_3 & \geq & 2. \end{array}$$

6. Let Ω be the the shaded plane region in the figure to the right. Restate using a system of inequalities $A\mathbf{x} \leq \mathbf{b}$ first (a) the constraints of the **primal** problem and then, (b) state the **dual** of the LP-problem 653005



$$\begin{cases} \max & z = 12x_1 + 17x_2, \\ & (x_1, x_2) \in \Omega. \end{cases}$$

7. Let Ω be the the shaded plane region in the figure to the right. Restate using a system of inequalities $A\mathbf{x} \leq \mathbf{b}$ first (a) the constraints of the **primal** problem and then, (b) state the **dual** of the LP-problem 30ace8



$$\begin{cases} \max & z = 3x_1 + 2x_2, \\ & (x_1, x_2) \in \Omega. \end{cases}$$

8. Find the dual of the LP problem:

$$\begin{array}{rrcrcrcl} \max z = & 6y_1 & + & 10y_2 & + & 12y_3 & & \\ \text{s.t.} & 3y_1 & + & y_2 & + & y_3 & \leq & 20 \\ & y_1 & + & 3y_2 & + & 4y_3 & \leq & 70 \\ & y_1 \geq 0, & & y_2 \geq 0, & & y_3 \geq 0. & & \end{array}$$

Solve it then by checking the value of the objective function in all corner points.

9. Resolve the LP to the right in any of the following ways: 22b0cb

- by solving the primal graphically (easy);
- by formulating and then solving the Dual (easy);
- by solving the primal via the M-method (quite difficult).

$$\begin{cases} \min & z = \begin{bmatrix} 60 & 90 \end{bmatrix} \mathbf{x}, \ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & \begin{bmatrix} 4 & 1 \\ 1 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 80 \\ 50 \\ 80 \end{bmatrix} \end{cases}$$

10. Resolve the min LP problem to the right by formulating and then solving its dual by the Simplex method. Indicate clearly the min value z^* of the objective function as well as the point x^* at which that min of the primal problem is attained. Which constraints are active at optimum? 22b0cc

$$\begin{cases} \min & z = \begin{bmatrix} 40 & 30 \end{bmatrix} \mathbf{x}, \ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 330 \\ 120 \\ 310 \end{bmatrix} \end{cases}$$

11. Formulate the dual problem for the LP: c4b72a

$$\begin{array}{rrcrcrcl} \min w = & 30y_1 & + & 12y_2 & + & 8y_3 & & \\ \text{s. t.} & 3y_1 & + & y_2 & + & y_3 & \geq & 6 \\ & 2y_1 & + & y_2 & & & \geq & 5 \\ & y_1 \geq 0, & & y_2 \geq 0, & & y_3 \geq 0. & & \end{array}$$

Then find the optimal value either graphically or using the Simplex Method.

12. Use the Simplex Method to solve the LP: 5307e5

$$\begin{array}{rcrcrcrcl} \max z & = & 3x_1 & + & x_2 & + & 2x_3 & + & 5x_4 & & \\ \text{s.t.} & & 2x_1 & + & x_2 & + & 3x_3 & + & 2x_4 & \leq & 120 \\ & & x_1 & + & x_2 & + & 2x_3 & + & 2x_4 & \leq & 90 \\ & & x_1 \geq 0, & & x_2 \geq 0, & & x_3 \geq 0, & & x_4 \geq 0. & & \end{array}$$

13. Solve (Big M, graphic or via the dual) the minimization problem b28a56

$$\begin{array}{rcrcrcl} \min z = & 3x_1 & + & 4x_2 & & \\ \text{s.t.} & x_1 & + & x_2 & \geq & 60 \\ & x_1 & + & 2x_2 & \geq & 72 \\ & \mathbf{x} \geq \mathbf{0}. & & & & \end{array}$$

14. Formulate the dual (D) of the LP of Exercise 5 and *solve it graphically.* c20c6e

15. Consider the LP to the right, call it (P): 800bbe

$$\begin{array}{rcrcrcl} \min z = & 10x_1 & + & 21x_2, & \mathbf{x} \geq \mathbf{0} & \\ \text{s.t.} & 2x_1 & + & 3x_2 & \geq & 6 \\ & 5x_1 & + & 7x_2 & \leq & 35 \end{array}$$

   (a) Find the dual (D) LP of the primal problem (P).

   (b) Find the optimum z^* by solving either (P) or (D).

16. Solve the LP problem below by formulating the dual, sketch the feasible region and the solve it graphically: 952c10

$$\begin{array}{rcrcrcrcl} \max z = & -x_1 & + & x_2 & - & x_3, & \mathbf{x} \geq \mathbf{0} & & \\ \text{s.t.} & 2x_1 & + & 2x_2 & + & x_3 & \leq & 10 \\ & x_1 & & & + & 2x_3 & \geq & 5 \end{array}$$

17. Formulate the **dual** of the LP problem to the right. Graph the feasible region of that dual problem and identify the point where the optimum is attained. Finally, compute the common optimal value of both the primal and the dual LP. f86afe

$$\max \quad z = 5x_1 + 3x_2 + 6x_3,\ \mathbf{x} \geq \mathbf{0}, \qquad \text{s.t.} \begin{cases} 2x_1 - 3x_2 + 5x_3 \geq 30 \\ 4x_1 + 2x_2 + 3x_3 \leq 240 \end{cases}$$

18. The LP below obtains its optimum at $x^* = (22, 0, 0, 1)$. 77e64e

$$\begin{array}{rcrcrcrcrcl} \min z = & 11x_1 & + & 17x_2 & - & 25x_3 & + & 12x_4, & \mathbf{x} \geq \mathbf{0} & \\ \text{s.t.} & 2x_1 & - & x_2 & + & x_3 & + & x_4 & \geq & 45 \\ & x_1 & + & 5x_2 & + & 3x_3 & - & 7x_4 & \leq & 80 \\ & 3x_1 & + & x_2 & + & 5x_3 & + & x_4 & = & 67 \end{array}$$

Formulate and solve its dual LP. State explicitly the values of the dual variables at optimum.

Sensitivity

1. TriCo manufactures three kind of products, p_1, p_2 and p_3, each requiring two stages of processing. The required processing time per stage, time available and unit profit are given in the table below: 48cfd4

	p_1	p_2	p_3	Available hours
Stage 1	7	1	2	11
Stage 2	3	2	1	5
Profit [$/u]	16	3	5	

Optimizing the product mix for maximal profit, it was concluded that TriCo should produce only goods p_1 and p_3, while p_2 (which brings in just \$3 per sold unit) is not worth to produce.

   (a) Find the quantities produced as well as TriCo's profit at optimum.

   (b) How much must the unit profit of p_2 increase before it becomes worth to produce?

(c) How much can the time available in Stage 2 vary before a change of plan becomes necessary?

2. Consider the LP below solved by the Simplex method with final Tableau given to the right: 484da2

$$\begin{array}{rcrcrcrcl} \max z & = & 4x_1 & + & x_2 & + & 11x_3, & \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & & 3x_1 & + & x_2 & + & 7x_3 & \leq & 72, \\ & & 2x_1 & + & x_2 & + & 6x_3 & \leq & 60. \end{array} \quad \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	s_1	s_2	rhs
z	0	3/4	0	1/2	5/4	111
x_1	1	−1/4	0	3/2	−7/4	3
x_3	0	1/4	1	−1/2	3/4	9

How much can the optimal value z^* be improved, without changing the basis, by increasing the resources in the rhs from their initial values of $b_1 = 72$ and $b_2 = 60$?

3. The LP below, solved by the Simplex Method, terminates with a final Tableau given to the right: 484da4

$$\begin{array}{rcrcrcrcl} \max z & = & 4x_1 & + & x_2 & + & 11x_3, & \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & & 3x_1 & + & x_2 & + & 7x_3 & \leq & 72, \\ & & 2x_1 & + & x_2 & + & 6x_3 & \leq & 60. \end{array} \quad \overset{SM}{\rightsquigarrow}$$

$\perp$	x_1	x_2	x_3	s_1	s_2	rhs
z	0	3/4	0	1/2	5/4	111
x_1	1	−1/4	0	3/2	−7/4	3
x_3	0	1/4	1	−1/2	3/4	9

In what range can the third cost coefficient c_3 in the objective function vary from its current value of 11 without having to change the basis? In what range will the optimal solution z^* vary then?

4. Consider the LP below solved by the Simplex method with final Tableau given to the right: 7bad82

$$\begin{array}{rcrcrcrcl} \max z = & 6x_1 & + & 10x_2 & + & 12x_3 & & \\ \text{s.t.} & 3x_1 & + & x_2 & + & x_3 & \leq & 20 \\ & x_1 & + & 3x_2 & + & 4x_3 & \leq & 70 \\ & x_1 \geq 0, & & x_2 \geq 0, & & x_3 \geq 0. & & \end{array} \quad \overset{SM}{\rightsquigarrow}$$

$\perp$	x_1	x_2	x_3	s_1	s_2	rhs
z	8	0	0	4	2	220
x_2	11	1	0	4	−1	10
x_3	−8	0	1	−3	1	10

(a) How much a unit increase of each of the right hand sides $b_1 = 20$ and $b_2 = 70$, one at a time, betters the optimal value z^*?

(b) What is the maximal profit z^* which can be achieved by increasing b_1 while leaving the basis unchanged?

5. Consider the LP: 71483e

$$\begin{array}{rcrcrcrcl} \max z & = & 3x_1 & + & x_2 & + & 2x_3 & & \\ \text{s.t.} & & 2x_1 & - & x_2 & + & 3x_3 & \leq & 18 \\ & & -x_1 & + & x_2 & - & 2x_3 & \leq & 12 \end{array}$$

(a) Find the optimum z^* using the Simplex Method and give point $\mathbf{x}^*$ at which it is attained.

(b) If you could increase by 1 either resource b_1 or b_2 in the rhs, which one will you choose, $b_1 : 18 \nearrow 19$, or $b_2 : 12 \nearrow 13$? By how much will the optimum z^* be improved?

(Hint: What are the rates of change $y_i = \dfrac{\partial z^*}{\partial b_i}$ w.r.t. b_i, and where do you find these in the final Tableau?)

6. ManuCo can manufacture six kind of goods in quantities $\mathbf{x}^T = (x_1, \cdots, x_6) \geq \mathbf{0}$, each reguiring three stages of processing. The unit profit $\mathbf{c}^T$ in \$/unit, the required hours/unit a_{ij} per stage and the available capacity $\mathbf{b}$ in hours are given in the table below: 4c5aa9

	p_1	p_2	p_3	p_4	p_5	p_6	$\mathbf{b}$
stage 1	1	1	1	2	2	6	216
stage 2	4	5	1	4	3	4	432
stage 3	2	0	5	2	1	0	648
$\mathbf{c}^T$	12	9	9	15	13	19	

A group of JU-students helps to optimize ManuCo's product mix $\mathbf{x}$. They formulate and solve the corresponding LP-problem by the Simplex Method obtaining a final Tableau at max as follows:

$\perp$	x_1	x_2	x_3	x_4	x_5	x_6	s_1	s_2	s_3	rhs
z	0	13/4	0	1/2	0	9	7/2	7/4	3/4	1998
x_5	0	1/8	0	3/4	1	4	3/4	−1/8	−1/8	27
x_1	1	31/24	0	5/12	0	−2	−7/12	3/8	1/24	63
x_3	0	−13/24	1	1/12	0	0	1/12	−1/8	5/24	99

(a) What is the optmal product mix for ManuCo, i.e., which goods are produced, and in what quantities?

(b) How much will the optimum profit of ManuCo drop, if, due to downtime, the available time b_2 in stage 2 is reduced from $432\searrow 428$ hours?

(c) How much is the unit profit c_5 of product 5 (originally, 13\$/u) allowed to drop before ManuCo needs to reconsider its product mix in order to stay at optimum, i.e., to change the basis in the LP?

7. HexoCo produces $\{p_i\}_1^6$ in quantities $\mathbf{x} \in \mathbb{R}^6_+$, each reguiring three stages of processing. The unit profits $\mathbf{c}$, the required hours/unit a_{ij} per stage and the available capacity $\mathbf{b}$ in hours are given in the table below: 4c5aaa

	p_1	p_2	p_3	p_4	p_5	p_6	$\mathbf{b}$
stage 1	1	1	1	2	2	6	216
stage 2	4	5	1	4	3	4	432
stage 3	2	0	5	2	1	0	648
$\mathbf{c}^T$ [\$/u]	12	9	9	15	13	19	

In order to find an optimal product mix $\mathbf{x}$ which maximizes the profit, a LP-problem was formulated. Its solution results in a final Simplex Tableau at max as follows:

$\perp$	x_1	x_2	x_3	x_4	x_5	x_6	s_1	s_2	s_3	rhs
z	0	13/4	0	1/2	0	9	7/2	7/4	3/4	1998
x_5	0	1/8	0	3/4	1	4	3/4	−1/8	−1/8	27
x_1	1	31/24	0	5/12	0	−2	−7/12	3/8	1/24	63
x_3	0	−13/24	1	1/12	0	0	1/12	−1/8	5/24	99

(a) State the max profit for HexaCo and the product mix (goods, quantities) at which it is attained.

(b) To what extent does HexaCo uses its production capacities at optimum? In which one of the processing stages is an hour of downtime most harmful? How much would the profit drop if it actually occurs?

(c) How much may the unit profit c_1 of product 1 (originally at 12 \$/u) drop before HexoCo has to reconsider its product mix (i.e., to change the basis in the LP) in order to stay at optimum?

8. The LP problem below, solved with the Simplex Method, has the optimal Tableau to the right: 4ca2da

$$\begin{aligned} \min z = \quad & -2x_1 - 2x_2 - 3x_3, \quad \mathbf{x} \geq 0, \\ \text{s. t.} \quad & x_1 + 3x_2 + x_3 \leq 9 \\ & x_1 + x_2 + 2x_3 \leq 15 \\ & 3x_1 + 2x_2 + x_3 \leq 72 \end{aligned} \quad \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	s_1	s_2	s_3	rhs
z	0	−2	0	−1	−1	0	−24
x_1	1	5	0	2	−1	0	3
x_3	0	−2	1	−1	1	0	6
s_3	0	−11	0	−5	2	1	57

(a) Which constraints are active in optimum?

(b) Find the ranges in which the right hand sides of the active constraints can vary, one at a time, without change of basis.

(c) What is the lowest possible optimal value z^* which can be obtained by perturbing the rhs of just one active constraint so that the basis remains unchanged?

9. ProdCo manufactures four different products and tries to maximize its profit. The products require three stages of processing, an ProdCo's available capacities for stages 1, 2 and 3 are 910, 800 and 480 hours correspondingly. The initial Simplex tableau of the resulting LP in standard form is given below: d6a384

$\perp$	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs
z	−90	−160	−40	−100	0	0	0	0
s_1	7	8	3	5	1	0	0	910
s_2	5	4	8	5	0	1	0	800
s_3	2	8	4	2	0	0	1	480

The coefficients of the objective function define the profit per unit and the coefficients in the left hand side of the restrictions — the processing time in hours per unit required in the corresponding stage. After performing the steps prescribed by the Simplex algorithm, the following optimal is obtained:

$\perp$	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs
z	10	0	130	0	0	15	25/2	18000
s_1	2	0	−11/2	0	1	−3/4	−5/8	10
x_4	1	0	3/2	1	0	1/4	−1/8	140
x_2	0	1	1/8	0	0	−1/16	5/32	25

(a) State the algebraic formulation of the LP and the produced quantities $\mathbf{x}^*$ for optimal profit z^*.

(b) How much an additional hour of capacity in either stage 1,2 or 3 would affect ProdCo's profit?

(c) At what minimal level of profit/unit c_3 would product 3 become worth producing?

(d) Provided base does not change, find the drop of the profit z^*, if, due to disruptions, the stage 2 capacity b_2 shrinks from 800 down to 400 h, and then to just 200 h.

(e) In what range can the unit profit c_4 vary with no change in the produced quantities?

10. The LP problem below, solved with the Simplex Method, has the optimal Tableau to the right: 4421d5

$$\begin{array}{ll} \max & z = 3x_1 + 2x_2 + x_3 \\ \text{s.t.} & \begin{cases} x_1 + x_2 + x_3 & \leq 120 \\ 2x_1 + 3x_2 - x_3 & \leq \ 90 \\ x_1, x_2, x_3 & \geq 0. \end{cases} \end{array} \quad \rightsquigarrow$$

	x_1	x_2	x_3	s_1	s_2	$\hat{\mathbf{b}}$
z	0	5/3	0	5/3	2/3	**260***
x_3	0	−1/3	1	2/3	−1/3	50
x_1	1	4/3	0	1/3	1/3	70

State the max value z^* of the objective function as well as the point x^* at which it is attained. Which constraints are active at optimum? Find also the ranges the following parameters can vary, *one at a time, basis unchanged*, in the LP exerc.

(a) Compute the ranges for b_1 of b_2, currently $\mathbf{b}^T = (120, 90)$, of the right hand side.

(b) Ditto for the coefficients c_1 and c_3 of the objective function. Which is the min value for c_2 for which the variable x_2 enters basis (i.e production of that product becomes profitable)?

(c) With how much will the optimal z^* increase if the supply b_1 were raised from 120 → 121?

11. The LP problem below, solved with the Simplex Method, has the optimal Tableau to the right: de76bb

$$\begin{array}{rrcrcrcr} \max z = & 12x_1 & + & 20x_2 & + & 15x_3 & & \\ \text{s. t.} & 2x_1 & + & 5x_2 & + & 3x_3 & \leq & 50 \\ & x_1 & + & x_2 & + & x_3 & \leq & 12 \\ & & & x_2 & & & \leq & 5 \\ & x_1 \geq 0, & & x_2 \geq 0, & & x_3 \geq 0. & & \end{array} \quad \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	s_1	s_2	s_3	$\hat{\mathbf{b}}$
z	3	0	0	0	15	5	205
s_1	−1	0	0	1	−3	−2	4
x_3	1	0	1	0	1	−1	7
x_2	0	1	0	0	0	1	5

(a) Which constraints are active in optimum? Which one of the components 50, 12 and 5 in the right hand side $\mathbf{b}$, increased by one, improves the optimal value z^* the most and by how much?

(b) Find the ranges in which the rhs $b_2 = 12$ can vary without change of basis.

(c) In which ranges can the objective function coefficient $c_3 = 15$ vary without change of basis?

12. The LP below is solved by the Simplex method with final Tableau as shown to the right: 32d009

$$\max \quad z = \begin{bmatrix} 70 & 22 & 94 & 53 & 9 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0},$$

$$\text{s.t.} \quad \begin{bmatrix} 23 & 7 & 31 & 18 & 3 \\ 16 & 5 & 22 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 33 \\ 23 \end{bmatrix} \quad \rightsquigarrow \{\text{SplxM}\} \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	rhs
z	1	0	3	4	0	1	3	102
x_5	3	0	1	−1	1	5	−7	4
x_2	2	1	4	3	0	−2	3	3

How much will the optimum $z^* = 102$ change if the right hand side of the second constraint is increased as much as possible from its current value of 23 before a change of basis becomes necessary?

13. The LP below is solved by the Simplex method with a final Tableau at max as follows: ee296d

$$\max \ z = \begin{bmatrix} 3 & 5 & 2 & 4 & 1 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0},$$

$$\text{s.t.} \begin{bmatrix} 4 & -1 & 1 & 3 & -1 \\ 1 & 2 & -1 & 2 & 1 \\ -1 & 1 & 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 42 \\ 36 \\ 30 \end{bmatrix} \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
z	0	0	0	1/3	11/3	2/3	2	5/3	150
x_1	1	0	0	11/12	−1/6	5/24	1/8	−1/24	12
x_2	0	1	0	5/12	5/6	−1/24	3/8	5/24	18
x_3	0	0	1	−1/4	1/2	1/8	−1/8	3/8	12

With how much can you improve the optimal profit z^* by increasing the second right hand side (the second resource) 36 as much as possible, without causing a change of basis?

14. PentaCo is after to maximize its profits by adjusting the quantities $x_1, \ldots, x_5$ of its five products, each undergoing three stages of processing. The profit per unit, the required processing times per stage, as well as the total time capacities available are as indicated in the table below: acf450

	P_1	P_2	P_3	P_4	P_5	Total capacity (h)
Profit (\$/unit)	11	9	4	7	5	
Time stage 1 (h/unit)	4	1	1	2	-1	521
Time stage 2 (h/unit)	2	3	1	1	3	1160
Time stage 3 (h/unit)	1	1	1	1	1	890

Keeping the ordering of the constraints, introducing a slack variable per constraint, and solving the LP by the Simplex method produces the final Tableau below, indicating a max profit of \$4008:

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
z	3	4/7	5/7	0	0	16/7	17/7	0	4008
x_4	2	6/7	4/7	1	0	3/7	1/7	0	389
x_5	0	5/7	1/7	0	1	$-1/7$	2/7	0	257
s_3	-1	$-4/7$	2/7	0	0	$-2/7$	$-3/7$	1	244

(a) Which products are produced at optimum and in what quantites? Which products are not profitable to produce? What are the least unit profits $\mathbf{c}_N$ at which these products can enter in production without incurring loss of profit?

(b) Expanding which of the three processing capacities by an hour gives best return? How much can PentaCo increase its profit by building up *only that* stage capacity without changing the basis?

(c) Find the widest range in which the profit c_5 of product 5 (currently $c_5 = \$5/\text{u}$) can vary without change of the basis (i.e, keeping the product mix of PentaCo)?

15. TillverkCo manufactures five items, sells them at unit profits $\mathbf{c}^T = (3, 1, 2, 5, 4)$, trying to max profit. Each item undergoes three processing operations of resource burning intensities in the matrix $A_{3\times 5}$ below. Finally, TillverkCo has available capacities for the three operations $\mathbf{b}^T = (1200, 2400, 1800)$. a9cf10

The corresponding max LP-problem is solved by the Simplex method with the final Tableau to the right:

$$\max \quad z = \begin{bmatrix} 3 & 1 & 2 & 5 & 4 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0},$$
$$\text{s.t.} \begin{bmatrix} 3 & 1 & 2 & 1 & 4 \\ 1 & 5 & 2 & 1 & 2 \\ 7 & 2 & 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 1200 \\ 2400 \\ 1800 \end{bmatrix} \rightsquigarrow$$

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
z	44/5	12/5	0	0	24/5	1/5	0	8/5	3120
x_3	2/5	1/5	1	0	7/5	3/5	0	$-1/5$	360
s_2	-2	4	0	0	-2	-1	1	0	1200
x_4	11/5	3/5	0	1	6/5	$-1/5$	0	2/5	480

How much are the unit profits of the items which actually get produced allowed to fall until a change of the production mix becomes necessary to mantain optimality? In other words, how much can the *coefficients* of the active variables in the objective function vary, one at a time, without a change of basis?

16. FünfCo fine tunes its product mix (i.e., the quantities $x_1, \ldots, x_5$ of its five products) by solving the LP: 5450f9

$$\begin{aligned} \max z \ &= 6x_1 + 6x_2 + x_3 + 4x_4 + 0x_5, \\ &= \mathbf{c}^T\mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \end{aligned} \qquad \text{s.t.} \begin{bmatrix} 4 & -1 & -1 & 3 & 1 \\ 1 & 2 & 1 & 2 & -1 \\ -1 & 1 & 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 14 \\ 12 \\ 10 \end{bmatrix}.$$

A group of JTH students solves it with the Simplex Method obtaining a final Tableau:

$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
z	0	0	3	4	0	1	3	1	60
x_1	1	0	$-1/6$	11/12	0	5/24	1/8	$-1/24$	4
x_2	0	1	5/6	5/12	0	$-1/24$	3/8	5/24	6
x_5	0	0	1/2	$-1/4$	1	1/8	$-1/8$	3/8	4

(a) State the max profit z^*, the point $\mathbf{x}^*$ at which it is attained and the constraints active at optimum.

(b) Find the range in which the cost coefficient c_5 (currently at 0) can vary so that item 5 still remains profitable to produce at optimum. In what range does the corresponding optimal profit z^* vary?

17. CincoCo fine tunes its product mix (the quantities $x_1, \ldots, x_5$ of its five products) by solving the LP: 5450fa

$$\begin{array}{ll} \max z & = 6x_1 + 6x_2 + x_3 + 4x_4 + 0x_5, \\ & = \mathbf{c}^T\mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \end{array} \quad \text{s.t.} \begin{bmatrix} 4 & -1 & -1 & 3 & 1 \\ 1 & 2 & 1 & 2 & -1 \\ -1 & 1 & 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 14 \\ 12 \\ 10 \end{bmatrix}.$$

A group of JTH students solves it with the Simplex Method obtaining a final Tableau:

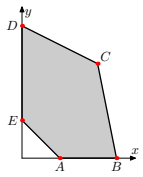
$\perp$	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
z	0	0	3	4	0	1	3	1	60
x_1	1	0	$-1/6$	$11/12$	0	$5/24$	$1/8$	$-1/24$	4
x_2	0	1	$5/6$	$5/12$	0	$-1/24$	$3/8$	$5/24$	6
x_5	0	0	$1/2$	$-1/4$	1	$1/8$	$-1/8$	$3/8$	4

(a) State the max profit z^*, the point $\mathbf{x}^*$ at which it is attained and the constraints active at optimum.

(b) If CincoCo wants to invest and build up its capacity, which one of the resources b_1, b_2 or b_3 in the right hand side should be increased to gain most in profit (*i.e., get most bang per invested buck*)? How much can that right hand side b_i increase until a change of basis becomes necessary to stay at optimum?

Parallelized

1. Describe the shadowed pentagon region with inequalities if its vertices are: 283713



$$A = (2, 0),\ B = (8, 0),\ C = (5,\ 6),\ D = (0, 8),\ E = (0, 2).$$

Hint: Can you deduce first the direction vectors of, and then, the normal to the lines bounding the region?

Octave & Excel

1. (`Excel+Matlab/Octave`) Find using Excel's Solver the optimal solution z^* and $\mathbf{x}^*$ for the LP: 5450f7

$$\begin{array}{ll} \max & z = \begin{bmatrix} 6 & 6 & 0 & 4 & 1 \end{bmatrix} \mathbf{x},\ \mathbf{x} \geq \mathbf{0}, \\ \text{s.t.} & \begin{bmatrix} 4 & -1 & 1 & 3 & -1 \\ 1 & 2 & -1 & 2 & 1 \\ -1 & 1 & 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \leq \begin{bmatrix} 14 \\ 12 \\ 10 \end{bmatrix} \end{array}$$

(a) Write the initial Simplex Tableau by introducing proper slack variables.

(b) Find the basis matrix B, its innverse B^{-1}, and reconstruct the last Simplex tableau at optimum by filling the proper places in the following Matlab code::

```
format rat;
c = [ 6 6 0 4 1 ]';                 % Def the param of the problem
A = [ 4 -1  1  3  -1
      1  2 -1  2   1
     -1  1  2 -1   2]; [m,n] = size(A);
b = [14 12 10 ]';

jB = [...]; B = A(:,jB);     % <- FILL indices of the basic cols of A and form B
jN = [...]; N = A(:,jN);     % <- FILL indices of non-basic cols of A and form N
invB = inv(B)                % Find the inverse of B
T0 = [A,eye(3),b]            % Glue the constraint rows of the initial tableau
Y  = c(jB)'*invB             % Compute the dual vars = -RC of the slack
CN = c(jN)'-Y*N              % Compute the reduced costs for non-basic vars
Zmax = Y*b                   % Compute z*
```

```
Tend = invB*T0              % The constraint rows of the last Tableau
Zrow = zeros(1,n);          % Fill Z-row with zeroes
Zrow(jN)=-CN;               % Fill the reduced costs for the non-basic vars
Zrow = [Zrow,Y,Zmax]        % glue dual vars and the value at opt z*
Tend = [Zrow;Tend]          % <-- Voilà, le Tableau lui même!
```